Speculative Decoding

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1 Foundamental Math

1.1 KL divergence

$D_{\mathrm{KL}}(P \,\|\, Q)$ measures how much a distribution Q diverges from a reference distribution P: the expectation, under P, of the log-ratio $\log \frac{P(x)}{Q(x)}$.

$$D_{\mathrm{KL}}(P \,\|\, Q) \;=\; \sum_x P(x) \log \frac{P(x)}{Q(x)}.$$

It is non-negative, equals 0 iff $P = Q$, and is asymmetric, so it is a divergence rather than a true distance. In ML it is used to quantify how far a model's predicted distribution Q is from the true distribution P.

1.2 Cross entropy

$H(P, Q)$ is the expected $-\log Q(x)$ under the true distribution P—intuitively, how "surprised" samples drawn from P are under the model's distribution Q:

$$H(P, Q) \;=\; -\sum_x P(x) \log Q(x).$$

Writing the entropy of P as $H(P) = -\sum_x P(x) \log P(x)$, the three quantities satisfy

$$H(P, Q) \;=\; H(P) + D_{\mathrm{KL}}(P \,\|\, Q),$$

i.e. cross entropy is the true entropy plus the extra cost of using Q instead of P. Since $H(P)$ does not depend on Q, minimizing cross entropy is equivalent to minimizing KL divergence, which is why cross entropy is the standard classification loss.

The one-hot case. In classification and next-token prediction the label is a single class/token, so the true distribution P is a one-hot vector: $P(c) = 1$ for the correct class c, 0 elsewhere. Then $H(P) = 0$, and:

$$H(P, Q) \;=\; D_{\mathrm{KL}}(P \,\|\, Q) \;=\; -\log Q(c) \;=\; \log \sum_j e^{z_j} - z_c,$$

where $Q = \mathrm{softmax}(z)$, z is logits.

2 Fast Speculative Decoding

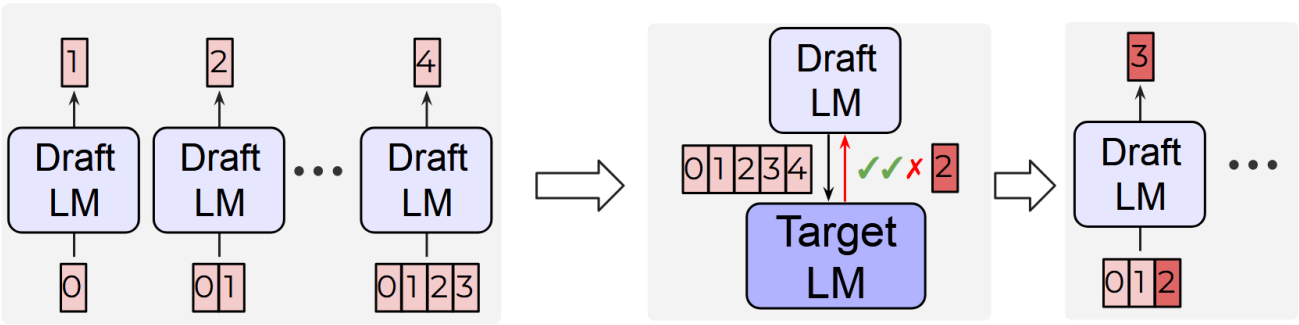


Figure 1: Speculative Deocoding Procedure.

2.1 Shift by 1 when verifying

When the target model verifies, input token ids `iL` are $[B, L]$ and output logits `oL` are $[B, L, V]$; `oL[i]` predicts the next token after `iL[i]`, so the label to verify is `iL[i+1]`. Hence the shift-by-1 pairs `oL[:-1]` with `iL[1:]`.

2.2 TODO

Might add after I fully grasp the math of paper